

1 Multivariate Calculus and Convexity

1. For each of the following functions, state if the function is homogeneous (if so, state its degree of homogeneity) and state whether it is homothetic. If your answer depends on the parameters, explain how it depends on the parameters. (Optional: prove that all these functions are quasiconcave.)

(a) $u(x_1, x_2, \dots, x_n) = \prod_{i=1}^n x_i^{\alpha_i}$

(b) $u(x_1, x_2, \dots, x_n) = \prod_{i=1}^n (x_i - b_i)^{\alpha_i}$

(c) $u(x_1, x_2, \dots, x_n) = \min\{a_1 x_1, \dots, a_n x_n\}$

2. Consider a production function $f : \mathbb{R}_+^n \rightarrow \mathbb{R}_+$. $f(x)$ is the amount of the single output good that can be produced using x_1 units of the first input, x_2 units of the second,..., x_n units of the n th input. Let us say f is additive if for any $x, y \in \mathbb{R}_+^n$, $f(x+y) \geq f(x) + f(y)$ (intuitively, if you can produce $f(x)$ with x and $f(y)$ with y , you should be able to produce at least $f(x+y)$ when you have both x and y). Say f is divisible if for any $t \in [0, 1]$, $x \in \mathbb{R}_+^n$, $f(tx) \geq tf(x)$ (intuitively, if you can produce $f(x)$ with x , you should at least be able produce half as much output when you only have half as much of every input). Prove that if a production function f is additive and divisible, then it is concave and homogeneous of degree 1.

3. In an economy with n goods with prices P_n , we define the aggregate price level P as:

$$P = A(P_1, P_2, \dots, P_n) = \left(\sum_{i=1}^n P_i^{1-\theta_i} \right)^\alpha$$

where $\theta_i \geq 0$ and $\theta_i \neq 1$ for all $i = 1, 2, \dots, n$, and $\alpha \in \mathbb{R}$.

(a) Is A homogeneous for all values of α and the θ_i 's?

From now on, consider the special case where $\theta_i = \theta \geq 0$ for all i .

(b) Loglinearize the definition equation of the price level around a point where all prices are the same: $P_1 = P_2 = \dots = P_n = P^*$ (and therefore $P = (n(P^*)^{1-\theta})^\alpha$). (Your answer needs to be independent of P^* .)

From now on, go back to the non-loglinearized aggregator.

- (c) For which value of α is A homogeneous of degree 1?

From now on, assume that α takes the value such that A is homogeneous of degree 1.

- (d) Show that A is quasi-concave.

8. Is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x^2 + y^2$ convex? Is the function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $g(x, y) = x^2 y^2$ convex?

9. Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x_1, x_2) := \frac{x_1 x_2}{x_1^2 + x_2^2 + 1}.$$

(a) For each $x \in \mathbb{R}^2$, calculate the partials $\partial f / \partial x_1(x)$ and $\partial f / \partial x_2(x)$.

(b) Use Theorem 1.7 to argue that f is differentiable at every $x \in \mathbb{R}^2$. What is the total derivative of f for each $x \in \mathbb{R}^2$?

$$\begin{aligned} a) \quad \frac{\partial f}{\partial x_1} &= \frac{(x_1^2 + x_2^2 + 1)(x_2) - (x_1 x_2)(2x_1)}{(x_1^2 + x_2^2 + 1)^2} & \text{Quot. } \frac{d}{dx} \frac{f(x)}{g(x)} &= \frac{g'(x)f(x) - f(x)g'(x)}{g(x)^2} \\ &= \frac{x_1^2 x_2 + x_2^3 + x_2 - 2x_1^2 x_2}{(x_1^2 + x_2^2 + 1)^2} & &= \frac{x_2(-x_1^2 + x_2^2 + 1)}{(x_1^2 + x_2^2 + 1)^2} \end{aligned}$$

$$\frac{\partial f}{\partial x_2} = \frac{(x_1^2 + x_2^2 + 1)(x_1) - (x_1 x_2)(2x_2)}{(x_1^2 + x_2^2 + 1)^2} = \frac{x_1(x_1^2 - x_2^2 + 1)}{(x_1^2 + x_2^2 + 1)^2}$$

$$b) \quad Df(x) = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \right] = \left[\frac{x_2(-x_1^2 + x_2^2 + 1)}{(x_1^2 + x_2^2 + 1)^2} \quad \frac{x_1(x_1^2 - x_2^2 + 1)}{(x_1^2 + x_2^2 + 1)^2} \right]$$

9. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x_1, x_2) := \frac{x_1 x_2}{x_1^2 + x_2^2 + 1}.$$

- (a) For each $x \in \mathbb{R}^2$, calculate the partials $\partial f / \partial x_1(x)$ and $\partial f / \partial x_2(x)$.
- (b) Use Theorem 1.7 to argue that f is differentiable at every $x \in \mathbb{R}^2$. What is the total derivative of f for each $x \in \mathbb{R}^2$?

$$a) \quad \frac{\partial f}{\partial x_1}$$

$$\text{Quot. } \frac{d}{dx} \frac{f(x)}{g(x)} = g'(x) f(x)$$

10. (a) Let $f(y) = ye^{-y^2/2}$. Calculate $\int_0^\infty f(y)y^2 dy$ using integration by parts. (You can use the fact that $\int_0^\infty f(y) dy = 1$.)
- (b) Let $S := \{(x_1, x_2) \in \mathbb{R}^2 : 1 \leq x_1 \leq 2, 0 \leq x_2 \leq x_1^3\}$ and function $f : S \rightarrow \mathbb{R}$ be defined as $f(x_1, x_2) = x_1 x_2$. Calculate the integral $\int_S f(x) dx$.
- (c) Let rectangle $S := \{(x_1, x_2) \in \mathbb{R}^2 : 1 \leq x_1 + x_2 \leq 5, -1 \leq x_1 - x_2 \leq 1\}$ and function $f : S \rightarrow \mathbb{R}$ be defined as $f(x_1, x_2) = x_1^2 + x_2^2 - 1$. Calculate the integral $\int_S f(x) dx$. (Hint: define

$$\begin{cases} u = x_1 - x_2 \\ v = x_1 + x_2 \end{cases}$$

and use the change of variables trick.)

11. Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ defined as

$$F(x) := \int_{-2x}^{x^2} (x^2 + e^t)^2 dt.$$

Find $F'(x)$ for each $x \in \mathbb{R}_+$.

12. Consider the system of linear equations:

$$\begin{cases} x_2 + x_1 y_1 + y_2^2 = 0 \\ x_1 - x_2 y_2 + y_1^2 = 0 \end{cases}$$

Use the implicit function theorem to argue that the system implicitly defines (y_1, y_2) as a function of (x_1, x_2) from an open ball around $(x_1^0, x_2^0) = (-1, 1)$ to an open ball around $(y_1^0, y_2^0) = (1, 0)$. What is the derivative of this implicit function at $(x_1^0, x_2^0) = (-1, 1)$?

13. Log-linearize the following equations.

(1) $\sum_{i=1}^n \alpha_i X_i = \sum_{i=1}^n X_i^{\beta_i}$ around $(X_1^*, \dots, X_n^*)$ satisfying this equation, where $\alpha_i > 0$, $\beta_i \neq 0$ for $i = 1, 2, \dots, n$.

(2) $\beta W = (1 - \alpha) \left(\frac{K}{N}\right)^\alpha$ around (W^*, K^*, N^*) satisfying this equation, where $\alpha \in (0, 1)$, $\beta > 0$.

(3) $P^{1-\varepsilon} = \alpha + (1 - \alpha) \left(\frac{Q}{R}\right)^{1-\varepsilon}$ around (P^*, Q^*, R^*) satisfying this equation, where $\varepsilon \neq 1$.

14. Suppose $x_{t+1} = \cos x_t$, $x_0 \in [0, 1]$.

(a) Show $x_\infty := \lim_{t \uparrow \infty} x_t$ exists in $(0, 1)$.

(b) Log-linearize the equation $x_{t+1} = \cos x_t$ around the point x_∞ , and for the approximated equation relating x_{t+1} to x_t , show $\lim_{t \uparrow \infty} x_t = x_\infty$.

15. In this exercise, you will show Taylor polynomials obey a special property that make them potentially good approximations for a function, namely that they uniquely share all derivatives of orders up to the degree of the polynomial with the function of interest at a particular point.

(a) Suppose $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = \frac{(x-c)^k}{k!}$ for $k \in \mathbb{Z}_{>0}$, $c \in \mathbb{R}$. Use induction to show

$$D^j g(x) = \begin{cases} \frac{(x-c)^{k-j}}{(k-j)!} & 0 \leq j < k \\ 1 & j = k \\ 0 & j > k \end{cases}$$

where D^j is the j -th order derivative, $j \in \mathbb{Z}_{\geq 0}$.

(b) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f \in C^n$, $n \in \mathbb{Z}_{\geq 0}$. Consider the polynomial $p(x) = \sum_{k=0}^n a_k \frac{(x-c)^k}{k!}$ where $a_k \in \mathbb{R}$. Show that for all $j = 0, \dots, n$, if $p^{(j)}(c) = f^{(j)}(c)$, then $a_j = f^{(j)}(c)$. Note the superscript (j) signifies the j -th order derivative.

2 Linear Algebra

1. Check whether the following sets are subspaces of the n -dimensional real vector space $\mathbb{R}^n$.¹

(a) Empty set $\emptyset$.

(b) $X := \{x \in \mathbb{R}^n : x = \alpha z, \text{ some } \alpha \in \mathbb{R}\}$, where $z \in \mathbb{R}^n$.

(c) (when $n = 3$) $S := \{(t - 2s, -s, t) : t, s \in \mathbb{R}\}$.

a) To show: $\emptyset \in \mathbb{R}^n$

Closed under addition

$\forall x, y \in \emptyset \Rightarrow x + y \in \emptyset$ trivially since
 $\emptyset = \{\}$ so $\forall x, y \in \emptyset$ is false

Closed under scalar multiplication

$\forall x \in \emptyset, \alpha \in \mathbb{R}, \alpha x \in \emptyset$ trivially
since $\emptyset = \{\}$ so $\forall x \in \emptyset$ is false

Existence of additive identity 0

$0 \notin \emptyset$ since $\emptyset = \{\}$

$\therefore \emptyset$ is not a vector subspace of $\mathbb{R}^n$

b) $X := \{x \in \mathbb{R}^n; x = \alpha z, \alpha \in \mathbb{R}, z \in \mathbb{R}^n\}$

Closed under Addition

$\forall x_1, x_2 \in X \Rightarrow x_1 + x_2 \in X$

$x_1 = \alpha_1 z_1, x_2 = \alpha_2 z_2 \quad \alpha_1, \alpha_2 \in \mathbb{R}$
 $z_1, z_2 \in \mathbb{R}^n$

$$\exists \alpha_3 \in \mathbb{R}, z_3 \in \mathbb{R}^n \text{ s.t.}$$

$$\alpha_1 z_1 + \alpha_2 z_2 = \alpha_3 z_3$$

Since $\mathbb{R}$ and $\mathbb{R}^n$ are vector spaces
and closed under addition

$$\therefore \alpha_3 z_3 = x_3 \in X \quad \text{and scalar multiplication}$$

Closed under Scalar Multiplication

$$\forall x = \alpha z, \alpha \in \mathbb{R} \text{ s.t. } z \in \mathbb{R}^n$$

$$\exists \beta \in \mathbb{R} \text{ s.t. } \beta x = \beta \alpha z$$

$\beta \alpha \in \mathbb{R}$ by closure of $\mathbb{R}$

$$\therefore \beta x \in X$$

Existence of additive identity 0

$$\exists 0 \in \mathbb{R}^n$$

$$\forall \alpha \in \mathbb{R}, \alpha 0 = 0$$

$$\therefore 0 \in X$$

all 3 conditions?
or 1th cond only?

All three conditions of a vector subspace are satisfied, therefore X is a vector subspace of $\mathbb{R}^n$

c) $S =: \{ (t - 2s, -s, t) : t, s \in \mathbb{R} \}$
Closure under addition

$$\forall (t_1 - 2s_1, -s_1, t_1), (t_2 - 2s_2, -s_2, t_2) \in S$$

$$\exists \lambda_1, \lambda_2 \in \mathbb{R}, t_3, s_3 \in \mathbb{R}$$

$$\lambda_1 (t_1 - 2s_1) + \lambda_2 (t_2 - 2s_2) = (t_3 - 2s_3)$$

$$\lambda_1 t_1 - 2\lambda_1 s_1 + \lambda_2 t_2 - 2\lambda_2 s_2 = t_3 - 2s_3$$

Grouping terms,

$$\lambda_1 t_1 + \lambda_2 t_2 = t_3 \quad -2\lambda_1 s_1 - 2\lambda_2 s_2 = -2s_3$$

substant? $-\lambda_1 s_1 - \lambda_2 s_2 = -s_3$

$\therefore$

$$\begin{aligned} \lambda_1 (t_1 - 2s_1, -s_1, t_1) + \lambda_2 (t_2 - 2s_2, -s_2, t_2) \\ = (t_3 - 2s_3, -s_3, t_3) \in S \end{aligned}$$

Since S is closed under linear combinations,
it is a vector subspace of $\mathbb{R}^3$

2. Show that the following $\|\cdot\|$ are valid norms in $\mathbb{R}^n$.

(a) $\|x\| := \max_{i=1}^n |x_i|.$

(b) $\|x\| := \sum_{i=1}^n |x_i|.$

3. (a) Suppose the vectors $v_1, v_2, \dots, v_n$ are linearly independent. Show that no one of these vectors can be expressed as a linear combination of the other $n - 1$ vectors.
- (b) Prove that the following claim is false: "Suppose the vectors $v_1, v_2, \dots, v_n$ are linearly dependent. Then any one of these vectors can be expressed as a linear combination of the other $n - 1$ vectors."

a) Given $v_1, \dots, v_n \in V$
are linearly independent,
$$c_1 v_1 + \dots + c_n v_n = 0 \quad (*)$$

has only trivial sol. $c_1 = \dots = c_n = 0$

For contradiction, pick $v_i \in V$

Assume
$$v_i = c_1 v_1 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_n v_n$$

Subtract v_i

$$0 = c_1 v_1 + \dots + c_{i-1} v_{i-1} + (-1) v_i + c_{i+1} v_{i+1} + \dots + c_n v_n$$

but (-1) is a valid c_i coefficient
and by $(*)$ $\sum_{i=1}^n c_i v_i = 0$ has
only trivial solution.

Since $c_i = -1 \neq 0$,
arrive at contradiction,
assumption must be false. $\square$

b) Given $v_1, \dots, v_n$ linearly dep.
We seek to disprove
$$v_n = c_1 v_1 + \dots + c_{n-1} v_{n-1}$$

for arbitrary v_n .

$0 = c_1 v_1 + \dots + c_{n-1} v_{n-1} + c_n v_n$ (*)
has nontrivial solutions
by linear dependence
such that not all $c_i = 0$.

counterexample:

$$\text{Let } v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Snt. V_1 & V_2 are linearly dep.,
but choosing V_1 as V_n ,
there are no C_2 s.t.

$$V_1 = C_2 V_2 \quad \text{since}$$

$$\forall C_2 \in \mathbb{R}, C_2 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Therefore V_1 cannot be
expressed as a linear
combination of V_2
& the statement is false. $\square$

4. Consider a vector space $(V, +, \cdot)$ over $\mathbb{R}$ with $\dim V = n$.

(a) Show that $v_1, v_2, \dots, v_n$ are a basis of V iff they are linearly independent.

(b) Show that $v_1, v_2, \dots, v_n$ are a basis of V iff any vector in V can be represented by a linear combination of them.

a) Basis requires $v_1, \dots, v_n = \text{span } V$
and $v_1, \dots, v_n$ to be linearly ind.

$\Rightarrow$ If $v_1, \dots, v_n$ is a basis of V , lin. ind. satisfied by def.

$\Leftarrow$ If $v_1, \dots, v_n$ lin. ind.,
 $\sum_{i=1}^n c_i v_i = 0 \Rightarrow \forall c_i = 0$ is the only sol.

Assume $v_1, \dots, v_n \neq \text{span } V$
s.t. $\exists x \in V$: x cannot be expressed as linear comb.
 $\sum_{i=1}^n c_i v_i$, thus $v_1, \dots, v_n, x$ is linearly ind.

Therefore $C_1V_1 + \dots + C_nV_n + C_{n+1}X = 0$

$\Rightarrow C_1 = \dots = C_n = C_{n+1} = 0$
is the only rational sol.

but $C_1V_1 + \dots + C_nV_n = -C_{n+1}X$
must be true s.t.

Divide
by 0?

$$\frac{1}{C_{n+1}} \sum_{i=1}^n C_i V_i = X$$

but X cannot be a
linear comb. of V_i

so C_{n+1} must be 0,

which means $V_1, \dots, V_n, X$

is a set of $n+1$ lin ind
vectors but $\dim V = n$

so there is no $\dim X$ can
exist in independent of $V_1, \dots, V_n$

Is this
Theorem?
if how
to prove?

5. Find non-zero 2×2 matrices A, B such that $AB = 0$.

$$5) \text{ Let } A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1+1 & 1-1 \\ -1+1 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

6. Determine the rank of the following matrices:

(a) $\begin{pmatrix} 1 & 3 & 4 \\ 2 & 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 3 & 0 & 0 \\ 2 & 4 & 0 & -1 \\ 1 & -1 & 2 & 2 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & -2 & -1 & 1 \\ 2 & 1 & 1 & 2 \\ -1 & 1 & -1 & -3 \\ -2 & -5 & -2 & 0 \end{pmatrix}$

6) a) $\begin{pmatrix} 1 & 3 & 4 \\ 2 & 0 & 1 \end{pmatrix} \xrightarrow{-2R_1} \begin{pmatrix} 1 & 3 & 4 \\ 0 & -6 & -7 \end{pmatrix}$ Rank 2

b) $\begin{pmatrix} 1 & 3 & 0 & 0 \\ 2 & 4 & 0 & -1 \\ 1 & -1 & 2 & 2 \end{pmatrix} \xrightarrow{-2R_1} \begin{pmatrix} 1 & 3 & 0 & 0 \\ 0 & -2 & 0 & -1 \\ 1 & -1 & 2 & 2 \end{pmatrix}$

$\xrightarrow{-R_1} \begin{pmatrix} 1 & 3 & 0 & 0 \\ 0 & -2 & 0 & -1 \\ 0 & -4 & 2 & 2 \end{pmatrix} \xrightarrow{-2R_2} \begin{pmatrix} 1 & 3 & 0 & 0 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 2 & 0 \end{pmatrix}$
Rank 3

$$c) \begin{pmatrix} 1 & -2 & -1 & 1 \\ 2 & 1 & 1 & 2 \\ -1 & 1 & -1 & -3 \\ -2 & -5 & -2 & 0 \end{pmatrix} \xrightarrow{-2R_1} \begin{pmatrix} 1 & -2 & -1 & 1 \\ 0 & 5 & 3 & 0 \\ -1 & 1 & -1 & -3 \\ -2 & -5 & -2 & 0 \end{pmatrix}$$

$$\xrightarrow{+R_1} \begin{pmatrix} 1 & -2 & -1 & 1 \\ 0 & 5 & 3 & 0 \\ 0 & -1 & -2 & -2 \\ -2 & -5 & -2 & 0 \end{pmatrix} \xrightarrow{+\frac{1}{5}R_2} \begin{pmatrix} 1 & -2 & -1 & 1 \\ 0 & 5 & 3 & 0 \\ 0 & 0 & -\frac{12}{5} & -2 \\ -2 & -5 & -2 & 0 \end{pmatrix}$$

$$\xrightarrow{+2R_1} \begin{pmatrix} 1 & -2 & -1 & 1 \\ 0 & 5 & 3 & 0 \\ 0 & 0 & -\frac{12}{5} & -2 \\ 0 & -1 & -4 & 2 \end{pmatrix} \xrightarrow{+\frac{4}{5}R_2} \begin{pmatrix} 1 & -2 & -1 & 1 \\ 0 & 5 & 3 & 0 \\ 0 & 0 & -\frac{12}{5} & -2 \\ 0 & 0 & \frac{7}{5} & 2 \end{pmatrix}$$

$\frac{27}{5} = 5\frac{2}{5}$

$$\xrightarrow{+R_3} \begin{pmatrix} 1 & -2 & -1 & 1 \\ 0 & 5 & 3 & 0 \\ 0 & 0 & -\frac{7}{5} & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{Rank 3}$$

7. Is it possible that the vectors v_1, v_2, v_3 are linearly dependent, but the vectors $v_1 + v_2, v_1 + v_3, v_2 + v_3$ are linearly independent?

Let $\{v_1, v_2, v_3\} = D$, a lin dep set

s.t. $\exists c_1, c_2, c_3 \in \mathbb{R}$ not all 0

$$\text{d. } c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \quad (*)$$

Assume $D' = \{v_1 + v_2, v_1 + v_3, v_2 + v_3\}$
are lin. ind. s.t.

$$c'_1(v_1 + v_2) + c'_2(v_1 + v_3) + c'_3(v_2 + v_3) = 0$$

has only the trivial solution $c'_1 = c'_2 = c'_3 = 0$

Then the following linear system cannot hold.

$$d_1(v_1 + v_2) = 0$$

$$d_1, d_2, d_3 \in \mathbb{R} \text{ not all } 0$$

$$d_2(v_1 + v_3) = 0$$

$$d_3(v_2 + v_3) = 0$$

but by $(*)$, $\exists c_1, c_2, c_3 \in \mathbb{R}$ not all 0

$$\text{s.t. } c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$$

multiply both sides by 2

$$2c_1 v_1 + 2c_2 v_2 + 2c_3 v_3 = 0$$

$$c_1 v_1 + c_1 v_1 + c_2 v_2 + c_2 v_2 + c_3 v_3 + c_3 v_3 = 0$$

stuck

7. Is it possible that the vectors v_1, v_2, v_3 are linearly dependent, but the vectors $v_1 + v_2, v_1 + v_3, v_2 + v_3$ are linearly independent?

Let $\{v_1, v_2, v_3\} = D$, a lin dep set

s.t. $\exists c_1, c_2, c_3 \in \mathbb{R}$ not all 0

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \quad (*)$$

To show possibility, ~

8. Let X be a $n \times k$ real matrix. Define projection matrix $P := X(X^\top X)^{-1}X^\top$ and orthogonal matrix $M := I_n - P$. (You can assume $(X^\top X)^{-1}$ exists.)
- (a) Show that P and M are idempotent.
 - (b) Show that $\text{tr}(P) = k$, $\text{tr}(M) = n - k$.

9. State whether each of the following statements is true or false. If it is false, please provide a counterexample.

- (a) No system of linear equations can have exactly k solutions for any $k \geq 2$.
- (b) If $Ax = 0$ has a solution, then $Ax = b$ has a solution;
- (c) If an $n \times n$ matrix A is full rank, then $Ax = b$ has a solution;
- (d) If an $n \times n$ matrix A has rank less than n , then $Ax = b$ has no solution;
- (e) If an $n \times n$ matrix A is full rank, all its eigenvalues are distinct.
- (f) Every diagonal real matrix has real eigenvalues.
- (g) An $n \times n$ matrix A has a zero eigenvalue if and only if it has rank less than n .

a) False. a linear system with free variables has infinite solutions.

Take $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$$Ax = b \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$x_1 = 1$$

$$x_2 = 1$$

x_3 free has infinite solutions

b) False,

$Ax = 0$ has the trivial solution $x = 0$,
but for $b \neq 0$, $Ax = b$ is not
guaranteed a solution

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- (f) Every diagonal real matrix has real eigenvalues.
- (g) An $n \times n$ matrix A has a zero eigenvalue if and only if it has rank less than n .

c) True, full rank $\Rightarrow$ invertible
so $x = A^{-1}b$

d) False, let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$ with rank 2
 $2 < 3 = n$
for $b = \begin{bmatrix} 1 \\ 1 \\ b_3 \end{bmatrix}$, $Ax = b$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ b_3 \end{bmatrix}$$

$x_1 = 1$
 $x_2 = 1$
 $x_3 = b_3$
has infinite solutions

e) False, let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, rank = 2

$$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{bmatrix} = (1-\lambda)^2 = 0$$

$$\lambda_1 = \lambda_2 = 1$$

9. State whether each of the following statements is true or false. If it is false, please provide a counterexample.

- (a) No system of linear equations can have exactly k solutions for any $k \geq 2$.
- (b) If $Ax = 0$ has a solution, then $Ax = b$ has a solution;
- (c) If an $n \times n$ matrix A is full rank, then $Ax = b$ has a solution;
- (d) If an $n \times n$ matrix A has rank less than n , then $Ax = b$ has no solution;
- (e) If an $n \times n$ matrix A is full rank, all its eigenvalues are distinct.
- (f) Every diagonal real matrix has real eigenvalues.
- (g) An $n \times n$ matrix A has a zero eigenvalue if and only if it has rank less than n .

f) True, for diag matrices, eigenvalues are given by the diagonal entries.
If all real, eigenvalues are real.

g) True, let $\lambda = 0$ be an eigenvalue of A & v the eigenvector,

$$Av = \lambda v = 0v = 0 \quad \text{so} \quad v \in \text{null}(A)$$

$$\det(A - \lambda I) = 0 \Rightarrow \det(A - 0) = \det(A) = 0$$

so A singular,

$$\text{thus } \text{rank}(A) < n$$

10. (a) Is the matrix

$$A = \begin{bmatrix} -1 & -2/3 \\ 3 & 5/3 \end{bmatrix}$$

diagonalizable in $\mathbb{C}$? Is it diagonalizable in $\mathbb{R}$? If so, diagonalize it, i.e. find a 2×2 invertible matrix P such that $P^{-1}AP = \Lambda$, where Λ is a diagonal matrix.

Consider a 2×2 real matrix as a vector in $\mathbb{R}^4$, and use d_2 on $\mathbb{R}^4$ to measure distance between two 2×2 real matrices. Find $\lim_{n \rightarrow \infty} A^n$.

- (b) Is the matrix

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

diagonalizable in $\mathbb{C}$? If so, diagonalize it.

11. Let $(V, +, \cdot, \|\cdot\|)$ be a normed vector space, and S_1 and S_2 two subsets of V . Define

$$S_1 + S_2 := \{x \in V \mid x = x_1 + x_2, x_1 \in S_1, x_2 \in S_2\}.$$

Consider the distance induced by the norm.

- (a) Assume S_1 and S_2 are bounded; is $S_1 + S_2$ bounded?
- (b) Assume S_1 and S_2 are closed; is $S_1 + S_2$ closed?
- (c) Assume S_1 and S_2 are compact; is $S_1 + S_2$ compact?

12. Solve the following systems of linear equations (find all real solutions). Write the answers in the vector form.

(a)

$$\begin{cases} x_1 + 2x_2 - x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 1 \\ 3x_1 + 5x_2 - 2x_3 = -1 \end{cases}$$

(b)

$$\begin{cases} x_1 - 2x_2 - x_3 = 1 \\ 2x_1 - 3x_2 + x_3 = 6 \\ 3x_1 - 5x_2 = 7 \\ x_1 + 5x_3 = 9 \end{cases}$$

a) $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 1 \\ 3 & 5 & -2 \end{bmatrix} \quad b = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$

by GE $\left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 2 & 2 & 1 & 1 \\ 3 & 5 & -2 & -1 \end{array} \right] \xrightarrow{-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & -2 & 3 & 3 \\ 3 & 5 & -2 & -1 \end{array} \right]$

$\xrightarrow{-3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & -2 & 3 & 3 \\ 0 & -1 & 1 & 2 \end{array} \right] \xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & -1 \\ 0 & -2 & 3 & 3 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array} \right]$

$$x_1 + 2x_2 - x_3 = -1$$

$$-2x_2 + 3x_3 = 3$$

$$-\frac{1}{2}x_3 = \frac{1}{2} \Rightarrow x_3 = -1$$

$$-2x_2 - 3 = 3 \Rightarrow -2x_2 = 6 \Rightarrow x_2 = -3$$

$$x = \begin{bmatrix} 4 \\ -3 \\ -1 \end{bmatrix}$$

$$x_1 - 6 + 1 = -1 \Rightarrow x_1 = 4$$

check

$$x_1 + 2x_2 - x_3 = -1$$

$$2x_1 + 2x_2 + x_3 = 1$$

$$3x_1 + 5x_2 - 2x_3 = -1$$

$$\Rightarrow 4 - 6 + 1 = -1$$

$$8 - 6 - 1 = 1$$

$$12 - 15 + 2 = -1$$

12. Solve the following systems of linear equations (find all real solutions). Write the answers in the vector form.

(a)

$$\begin{cases} x_1 + 2x_2 - x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 1 \\ 3x_1 + 5x_2 - 2x_3 = -1 \end{cases}$$

(b)

$$\begin{cases} x_1 - 2x_2 - x_3 = 1 \\ 2x_1 - 3x_2 + x_3 = 6 \\ 3x_1 - 5x_2 = 7 \\ x_1 + 5x_3 = 9 \end{cases}$$

12. Solve the following systems of linear equations (find all real solutions). Write the answers in the vector form.

(a)

$$\begin{cases} x_1 + 2x_2 - x_3 = -1 \\ 2x_1 + 2x_2 + x_3 = 1 \\ 3x_1 + 5x_2 - 2x_3 = -1 \end{cases}$$

(b)

$$\begin{cases} x_1 - 2x_2 - x_3 = 1 \\ 2x_1 - 3x_2 + x_3 = 6 \\ 3x_1 - 5x_2 = 7 \\ x_1 + 5x_3 = 9 \end{cases}$$

b) $x_1 = 9 - 5x_3$

$$3x_1 = 7 + 5x_2$$

$$x_1 = 7/3 + 5/3 x_2$$

$$9 - 5x_3 = 7/3 + 5/3 x_2$$

$$27 - 15x_3 - 7 = 5x_2$$

$$\frac{20 - 15x_3}{5} = x_2 = 4 - 3x_3$$

4 eq in 3 unknowns
one determined,
free variable.

$$x = \begin{bmatrix} 9 - 5x_3 \\ 4 - 3x_3 \\ x_3 \end{bmatrix}$$

part. sol.
let $x_3 = 1$

$$x_p = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

check

$$4 - 2 - 1 = 1$$

$$8 - 3 + 1 = 6$$

$$12 - 5 = 7$$

$$4 + 5 = 9$$

(c)

$$\begin{cases} x_1 + 2x_2 + 2x_4 = 6 \\ 3x_1 + 5x_2 - x_3 + 6x_4 = 17 \\ 2x_1 + 4x_2 + x_3 + 2x_4 = 12 \\ 2x_1 - 7x_3 + 11x_4 = 7 \end{cases}$$

(d)

$$\begin{cases} x_1 - 4x_2 - x_3 + 3x_4 = 2 \\ 2x_1 - 8x_2 + x_3 - 4x_4 = 9 \\ -x_1 + 4x_2 - 2x_3 + 5x_4 = -6 \end{cases}$$

13. For each of the matrices, check if it is diagonalizable over $\mathbb{C}$. Do the same over $\mathbb{R}$. If it is diagonalizable, then find invertible P and diagonal Λ so that $A = P\Lambda P^{-1}$. It may help to check your computation with your geometric understanding of eigenvalues and eigenvectors. Also, what is the definiteness of A (equivalently, the quadratic form associated with A), e.g. positive definite, positive semidefinite, etc.?

(a) $A = \begin{pmatrix} 0 & 1 \\ -1/2 & 0 \end{pmatrix}$

(b) $A = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$

(c) $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

(d) $A = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$

For (d), please also compute $\lim_{n \rightarrow \infty} A^n$, the limit taken with respect to the d_2 metric (considering A as an element of $\mathbb{R}^4$).